Aryan Kaul

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Research Focus

My research explores how to expand users' abilities by giving them more ways to access, understand, and control the systems and tools they use every day, using human-centered design to build and evaluate agentic and adaptive systems.

Education

Columbia University — Jan 2026 – May 2027
Master of Science, Computer Science, Thesis Track (GPA: 4.13) — *New York, NY*

- Research affiliations: Computer-Enabled Abilities Laboratory (Brian A. Smith), Computational Design Lab (Lydia B. Chilton), and Data, Agents & Processes Lab (research with Zhuo Zhang).
- Selected coursework: Applied Machine Learning; GenAI & Modern Deep Learning; Computational Complexity; HCI Graduate Seminar; Human-Centered Design and Innovation; Digital Storytelling with AI.

University of Maryland — Aug 2019 – May 2023
Bachelor of Science, Computer Science (GPA: 3.76/4.0); Minor in Business Analytics — *College Park, MD*

- Dean's List, 2019–2023; Office of Multi-Ethnic Student Education Academic Excellence Society.
- Selected coursework: Algorithms; Advanced Data Structures; Advanced Linear Algebra; Computer Systems; Cryptography; Data Science; Network Security; Programming Languages; Spectral Graph Theory.

Research Experience

Spoken URL Safety for Screen-Reader Users — Jan 2026 – Present
Computer-Enabled Abilities Laboratory; advised by Brian A. Smith — *Columbia University*

- Led end-to-end research and engineering on spoken URL safety for screen-reader users, spanning research framing, iterative system design, two IRB-approved studies, participant recruitment, analysis, and manuscript development.
- Engineered cross-platform prototypes spanning Android AccessibilityService/TalkBack, an iOS share-sheet app, browser extensions, and screen-reader plugins, refining warnings and inspection workflows from user feedback.
- Ran mixed-methods evaluations with 48 sighted and 14 blind participants, combining quantitative accuracy, workload, and interaction traces with manual qualitative coding and thematic analysis of interview transcripts.

Conversational Customization of Productivity Systems — 2026
Computational Design Lab; with Karthik Sreedhar and Lydia B. Chilton — *Columbia University*

- Built and studied a conversationally malleable email system as a design probe for how people create, refine, and oversee custom functionality in everyday software over multi-day use, including how features evolve with use.

Visual Diffs for Evaluating Structural Changes Made by Coding Agents — Jul 2026 – Present
Computational Design Lab; with Riya Sahni, Vivian Liu, and Lydia B. Chilton — *Columbia University*

- Built visual C4/Git diffs that track structural changes and architectural drift across agent turns, helping developers inspect whether coding-agent edits stay within scope or reshape a codebase unexpectedly.

Agents as Auditors: Detecting Malicious UI Flows via Trajectory-Level Analysis — May 2026
Data, Agents & Processes Lab; with Yuhang Chen and Zhuo Zhang — *Columbia University*

- Co-led a multi-agent VLM auditor for deceptive Android flows, combining screen-sequence reasoning, UIAutomator inspection, rule-based checks, benchmark construction, evaluation, and failure analysis across real apps.

Publications and Preprints

Riya Sahni, **Aryan Kaul**, Anushka Bhatnagar, Xuanming Zhang, Vivian Liu, Reya Vir, and Lydia B. Chilton. "Visual Diffs for Evaluating Structural Changes Made by Coding Agents." *arXiv preprint*, 2026.

Karthik Sreedhar, **Aryan Kaul**, and Lydia B. Chilton. "Conversational Customization of Productivity Systems." *arXiv preprint*, 2026.

Professional Experience

Software Engineer, Mobile Platforms Dec 2023 – Jan 2026
Amazon, Kindle *Seattle, WA*

- Modernized Kindle's Android and Fire Tablet platform (1M+ LOC) using MVVM architecture with Kotlin and Jetpack Compose, increasing engagement on large-form-factor devices by 2% for over 10 million readers.
- Spearheaded stability and build reliability by fixing 25+ crashes/freezes and triaging 200+ issues via stack traces, instrumenting SLIs in CloudWatch, and adding static-analysis gates across a 50+ developer organization.
- Scaled Kindle's design system as a top contributor, building 10+ accessible component APIs and design tokens; used Claude and other models by task to cut design drift 70% and save an estimated 100+ engineering weeks.
- Championed accessibility across native and embedded web surfaces, advising on semantics, focus, and contrast; resolved 90 tickets and reviewed 200+ PRs while mapping WCAG/ARIA requirements across implementations.

Software Development Engineer Intern May 2022 – Aug 2022
Amazon, Kindle *Seattle, WA*

- Prototyped and shipped Colored Bookmarks across Android and Fire Tablet, aligning UX flows and API surfaces with Design/PM while validating color-blind-safe tokens, screen-reader support, and cross-device behavior.
- Implemented the feature with Fragments/XML and test-driven development using JUnit, Robolectric, and Espresso, reaching 80%+ test coverage across bookmark interactions and regression cases.

Software Engineer, Full-Stack Jul 2023 – Nov 2023
Programmers.io *Remote*

- Built 8 ASP.NET Core MVC features spanning CRUD workflows, role-based auth, React/Chart.js dashboards, SQL Server tuning, and client-side caching, reducing negative customer reviews 40%.

Engineering Intern, Infrastructure Solutions Group May 2021 – Sep 2021
Dell Technologies *Hopkinton, MA*

- Built reporting and telemetry pipelines across 700+ programs with Atlassian REST APIs and JQL, replacing manual status decks with live executive dashboards for planning, risk, and OKR tracking.

Visiting Intern Jun 2018
CERN *Geneva, Switzerland*

- Selected for an in-person program led by Archana Sharma; studied LHC operations and detector systems including CMS, ALICE, and AMS, with technical seminars from researchers including Peter Jenni and Manjit Dosanjh.

Selected Talks, Workshops, and Projects

North East AI Agents Day May 2026
Jane Street *New York, NY*

- Presented trajectory-level VLM auditing work on deceptive mobile flows to researchers and industry practitioners, covering screen-sequence reasoning, automated UI inspection, benchmark design, evaluation, and failure analysis.

Global Accessibility Awareness Month Conference May 2025
Amazon *Seattle, WA*

- Presented to 60+ attendees on modernizing Kindle's mobile platform, covering staged framework migration, accessible design-system adoption, validation guardrails, and regression prevention across mobile and web.

Story I/O Vibe Coding Workshop Apr 2026
Columbia Digital Storytelling Lab *New York, NY*

- Co-facilitated an AI prototyping workshop, guiding artists, designers, and programmers from prompt decomposition through coding, debugging, and deployment using coding agents and AI-native development platforms.

Directed Reading Program Conference May 2021
University of Maryland *College Park, MD*

- Presented a proof sketch of Brouwer's Fixed Point Theorem via a retraction and fundamental-group contradiction, developed through semester-long work on topology in UMD's Directed Reading Program with Vlassis Mastrantonis.

Lions-Med Crowd May 2026
Human-Centered Design Workshop, PRI Innovation Forum *New York, NY*

- Presented a crowdsourced clinic-information prototype for real-time wait times, medication availability, prices, alerts, and peer confirmations to help navigate care during disruptions, crowds, shortages, or access barriers.

ReKindle: A Guided Friendship Experience 2026
Digital Storytelling with AI, Columbia University *New York, NY*

- Presented an immersive guided experience at Lincoln Center with an interdisciplinary team, prompting strangers to share stories about meaningful friendships and reconnect through technology-mediated reflection.

Teaching and Mentoring

Teaching Assistant, Discrete Structures Aug 2021 – May 2022
University of Maryland; with Justin Wyss–Gallifent and Clifford Bakalian *College Park, MD*

- Planned and delivered recitations, problem sets, and exam reviews; led weekly problem-solving sessions and office hours for a 40+ student section focused on proof writing, induction, and mathematical reasoning.
- Authored calibrated rubrics, proctored exams, and coordinated grading for a 650+ student cohort, aligning cross-section assessment, feedback standards, and academic-integrity policies.

Volunteer Reading Tutor Aug 2023 – Summer 2025
Reading Partners

- Tutored Title I elementary students one-on-one using a structured literacy curriculum, tracking comprehension and fluency while using interactive and STEM-based activities to sustain engagement.

Peer Mentor, Carillon Communities Aug 2020 – Dec 2020
University of Maryland *College Park, MD*

- Facilitated weekly IDEA101 Studio sessions for first-year students, coordinated mentoring plans with instructors, and applied design-thinking and peer-feedback methods through a parallel coaching seminar.

Pedagogical Training

Teaching Techniques for Computer Science (CMSC395), *with Jandelyn Plane, University of Maryland*
TA-focused training in effective teaching, inclusive classrooms, diversity and inclusion, and legal/ethical responsibilities.

Inquiry Approach to Teaching STEM (TLPL101), *Terrapin Teachers, University of Maryland*
Inquiry-based, student-centered STEM teaching through field experiences observing, planning, and teaching lessons.

Special Topics in Coaching: Carillon Peer Mentors (IDEA398C), *Carillon Communities, University of Maryland*
Two-credit practicum in design thinking, mentoring, leadership, and communication while co-teaching IDEA101.

Community and Service

HCI Seminar Co-Organizer Fall 2026
Columbia University *New York, NY*

- Co-organize Columbia's graduate HCI seminar with Tao Long, coordinating invited research talks, speaker visits, student and faculty meetings, and programming for the Columbia HCI community across labs and departments.

Citizen AI 2026 – Present
NSF-funded community-driven AI research initiative *New York, NY*

- Support community AI workshops and outreach while contributing technical input to a workshop-grounded knowledge base and conversational assistant exploring adoption, trust, and hallucination awareness.

Co-Founder, BigThink AI Jan 2021
University of Maryland *College Park, MD*

- Founded and grew a student AI organization to 100+ members, recruiting officers and organizing talks and discussions on AI applications across healthcare, finance, media, and education.

Volunteer Technology Tutor & Analyst Aug 2018
Enable India *India*

- Coached adults with low vision and motor impairments in Microsoft Office, email, web navigation, and accessibility tools; analyzed national-helpline call logs and produced weekly traffic summaries for staffing and outreach.

Technical Skills

Programming Languages: Python, Kotlin, Java, Swift, TypeScript, JavaScript, SQL
Machine Learning: PyTorch, scikit-learn, XGBoost, Neural Networks, Transformers, Model Training, Evaluation
AI Systems: LLMs, VLMs, RAG, Supervised Fine-Tuning, Multi-Agent Systems, Tool-Using Agents
AI Development: Claude Code, Codex, Cursor, Amazon Q Developer, Replit, Lovable, Figma Make
Mobile and Web: Android SDK, Fire OS, Jetpack Compose, Fragments/XML, AccessibilityService, UIAutomator, Robolectric, Espresso; iOS Accessibility APIs, XCUITest, VoiceOver; WebView, React, Chrome Extensions, WCAG 2.2, ARIA, APCA
Backend and Cloud: Supabase, Firebase, Docker, REST APIs, AWS, GCP, Vercel, CloudWatch, Git, Gradle, Jenkins